

Michael Ly, MPH

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SUMMARY

Data engineer building Snowflake Cortex AI and Streamlit tools for technical teams, non-technical users, and legacy systems. AI has moved my bottleneck from writing code to framing problems, breaking them into modular pieces, and designing around how people actually use the tool.

PROFESSIONAL EXPERIENCE

Data Engineer — Ludi Inc | Nashville, TN

Nov 2025 - Present

- Designed and shipped an AI-assisted DDL automation tool (Streamlit-in-Snowflake + Snowflake Cortex AI) that replaced a 2–4 hour manual SQL-authoring workflow with a sub-5-minute review-and-deploy process — cutting client onboarding time ~95% across 10+ enterprise healthcare clients, with deterministic safety rules and a mandatory human-review gate.
- Built a Streamlit-in-Snowflake app where Snowflake Cortex AI samples each column, drafts SQL cleaning logic, and shows a before-and-after of every value, with automated pre/post delta checks to catch data loss and a mandatory human-review gate; reviewers rarely needed to change its suggestions, replacing column-by-column manual review.
- Built event-driven ingestion on AWS: Workato and cron listeners watch partner SFTP paths and route files into S3, with tiered EventBridge dispatch (small files → Lambda, large files → Glue) and idempotent deduplication that eliminated silent payload-limit failures across SFTP → S3 → Snowflake.
- Shipped 6 production Streamlit-in-Snowflake apps used regularly by data and implementation teams, under Snowflake role-based access with audit logging: AI-assisted onboarding, flat-file onboarding, and view-repair tools (Cortex AI), backup-first guardrailed deletes with auto-generated reversal scripts (zero unrecoverable data-loss incidents), a pipeline run monitor, and a self-serve reference portal for analytics.
- Owned end-to-end client onboarding infrastructure on a 3-person team — multi-tenant SFTP on AWS EC2 (chrooted SSH, IP allowlisting), S3, Snowflake database provisioning, and Workato orchestration — for healthcare clients with strict compliance requirements.

Data Engineer II — Ellevation Education | Boston, MA

Oct 2024 – May 2025

- Led optimization and stabilization of production Airflow DAGs — reducing pipeline failure rate by 15% through improved error handling, retry logic, and on-call incident response for executive-facing dashboards driving 10% ARR growth.
- Architected and scaled AWS infrastructure (S3, EC2, SFTP, Redshift), reducing ETL runtimes and enabling scalable ingestion for high-volume partner data, saving 20% cloud utilization costs.

- Leveraging CI/CD workflows using Git to automate testing and deployment for reliable, production-ready data pipelining, leading to increased fault tolerance of infrastructure.

Data Engineer I — Ellevation Education

May 2023 – Oct 2024

- Created custom Python workflows by preparing raw partner data from AWS S3 and staging it in Redshift, ensuring accuracy and consistency for downstream analytics.
- Assisted in building and scheduling Airflow DAGs, gaining hands-on experience with pipeline orchestration and automation.

Data Engineer Intern — Brotman Baty Institute

Feb 2022 – Jan 2023

- Contributed to data engineering for the United States' first COVID-19 cases in Seattle, WA.
- Designed and optimized ETL processes using SQL and Python to aggregate survey and swab data from 50,000+ participants, generating over 2M records for epidemiological research.

TECHNICAL SKILLS

Snowflake AI & Apps: Snowflake Cortex AI, Streamlit in Snowflake, AI-assisted data prep, Human-in-the-loop review gates, RBAC, Audit logging

Languages & Tools: Python, SQL, Git, Pandas, Streamlit, Cortex AI, Data Modeling, ETL, SCD2

Cloud & Data Platforms: AWS (Lambda, S3, Glue, EventBridge, EC2, CloudWatch, Athena), Snowflake, Workato

Workflow & Orchestration: Schemachange, dbt, Apache Airflow, SFTP/S3 pipelines, Idempotent event-driven architecture

PROJECTS

[Incremental Data Pipelines with dbt](#) – Seattle, WA

August 2025 - September 2025

- Built incremental dbt models to process only new/updated records, reducing pipeline runtimes and warehouse costs by ~70%.
- Implemented schema validation tests and custom macros to enforce data integrity, detect schema drift, and ensure analytics-ready outputs.

[Comparing Kimball and Inmon Schema Performance in RedShift](#) – Seattle, WA

June 2025 - July 2025

- Benchmarked star vs. snowflake schema performance in AWS Redshift using real-world data volume.
- Created Python-based ETL pipelines for both schemas with data staged in S3 and loaded via COPY commands.
- Used EXPLAIN and query logs to identify storage and performance trade-offs, finding 37% faster reads with Kimball.

EDUCATION & CERTIFICATIONS

MPH in Epidemiology (Data Analytics) from UAB in Birmingham, AL

Status - Graduated

B.A. in Epidemiology from UW in Seattle, WA

Status - Graduated

AWS Certified Solutions Architect Associate

Status - Certified

AWS Certified Data Engineer – Associate

Status - Pending